
FOIS 2026 · EARLY CAREER SYMPOSIUM · FAST TALK

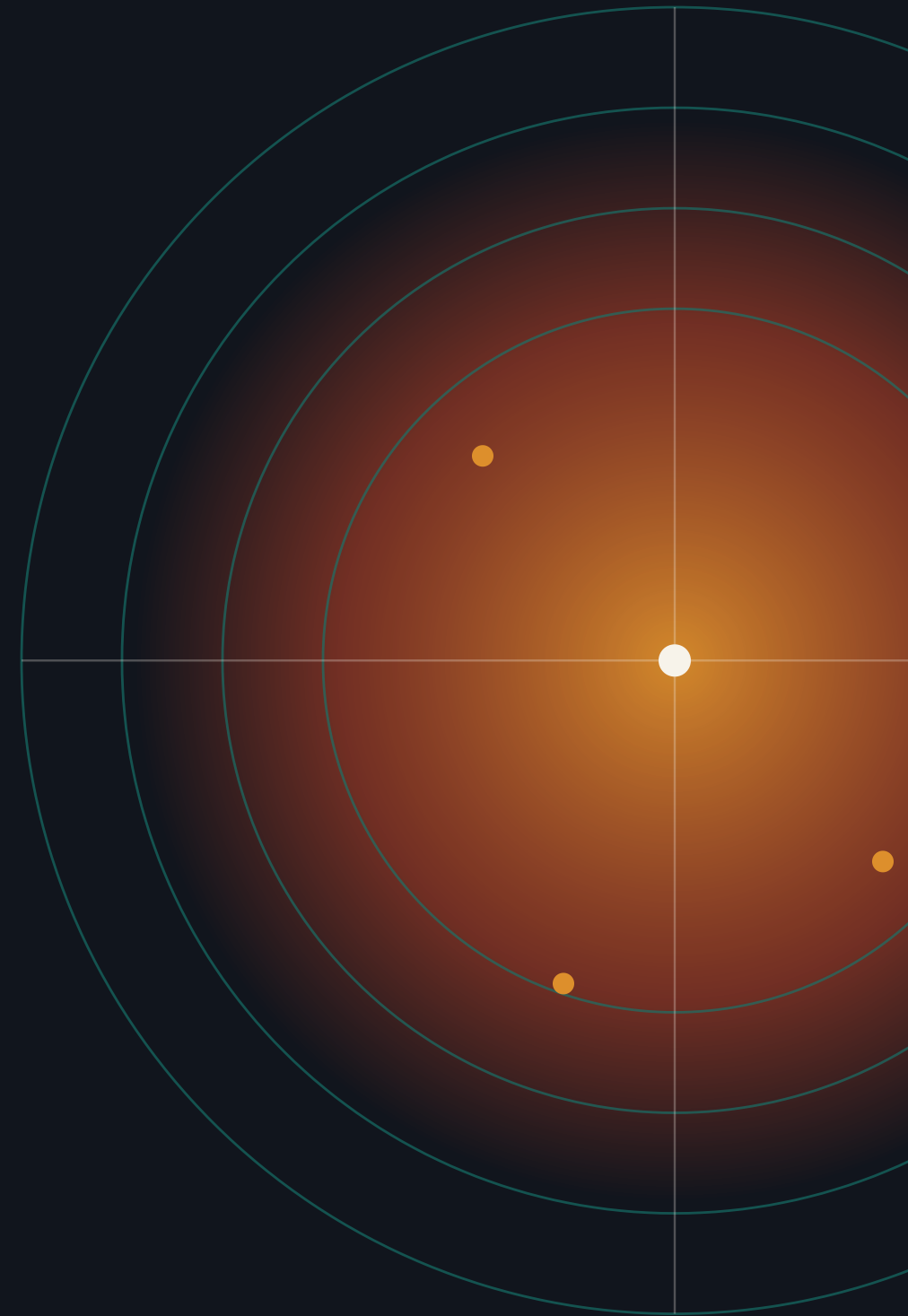
How Much Local Data Is Enough?

An ontology-grounded neurosymbolic research agenda
for transferability in forecasting

Rodrigo Huber Marques Moreira Mendes

Fernanda Baião · Reinaldo Castro Souza

Pontifícia Universidade Católica do Rio de Janeiro (PUC-Rio)



Brazil measures the sun in a *few places*.

- A handful of well-instrumented sites have years of clean records. **Vast regions have almost nothing.**
- The pragmatic hope is **transfer learning** [Pan & Yang 2010]: train where data is plentiful, leverage where it is not.

But leveraging is not epistemically neutral.

Leveraging presupposes the target is the *same kind of place* as the source.

TWO STATIONS MAY

share a latitude
but differ in cloud regime

TWO STATIONS MAY

share a biome
but differ in altitude

TWO STATIONS MAY

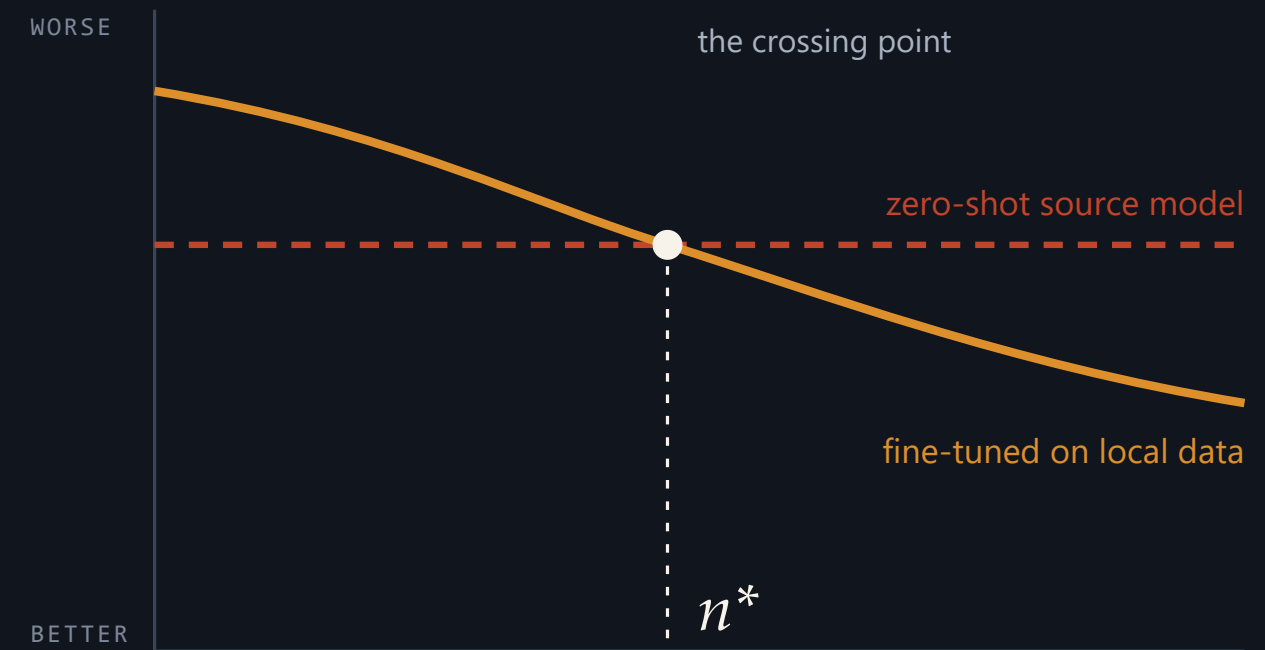
sit far apart
yet behave identically

Diverge along the dimensions that matter and transfer **degrades performance rather than improving it** — negative transfer. Which similarities warrant transfer is a question **formal ontology** was built to clarify.

How much local data is *enough*?

How much must be collected before **fine-tuning** beats
deploying the source model as-is?

Call it the site's **data budget**, n^* .



A network is mute about the scope of its own validity.

- It carries no account of **why** or **where** learned patterns should generalize. The neurosymbolic response is to couple the learner with an explicit symbolic layer that can be interrogated in human-legible terms [Garcez et al. 2023 · Kautz 2022].
- The promise is not only accuracy but **governance** — a model whose limits can be stated in plain terms, including where it should *not* be trusted.

*A symbolic gate is only as good as the **categories** it reasons with.*

Köppen climate types, biome schemes and bioclimatic distances **already are domain ontologies**: structured vocabularies for organizing the world. Grounding the gate in a **well-founded** one [Guizzardi 2005] is what makes its commitments semantically precise, inspectable and **independently auditable** [Tiddi et al. 2022] — rather than an ad-hoc feature set.

A Neurosymbolic Semantic Gate between the model and the place.

A symbolic decision layer, **still under development**, that would reason — in terms a domain scientist can read — about whether and how readily transfer is warranted.

IN

A trained source model
and a candidate target site

THE GATE

A cascade of five
symbolic criteria

OUT

A human-readable verdict
and a predicted n^*

APPROVED

WARNING

REJECTED

— a defeasible epistemic judgment, not a fact about the world.

Each criterion is a *distinct, testable hypothesis* about what licenses transfer.

CRITERION	NOTION OF “SAMENESS”	ONTOLOGICAL SOURCE
Radiative compatibility	Comparable top-of-atmosphere forcing	Astronomy / site geometry
Seasonal alignment	Matching shape of the annual cycle	Climatology
Climate identity	Same Köppen class	Köppen-Geiger classification
Ecological proximity	Near in biome / climate space	Whittaker biome scheme
Geographic nearness	Small great-circle distance	Spatial proximity

These are **distinct relations that often disagree**. Which notion of sameness, if any, actually carries predictive weight?

The gate will be measured against a *ground truth* that already exists.

20

BRAZILIAN SITES,
BUDGET ALREADY
ESTABLISHED

1

SOURCE SITE
GALEÃO, RIO DE JANEIRO

11–2,100

KM SPAN, TROPICAL RIO
BASIN TO THE
EQUATORIAL NORTH

H1 — ORDERING

Sites judged **more similar** require fewer local observations to adapt — a measurably lower n^* .

H2 — DISCRIMINATION

The five criteria are **not interchangeable** — the empirical record will tell them apart.

Controlled for target-site difficulty, so similarity is not confused with a site simply being easier to forecast.

Three overlapping work packages, *not* a sequence of phases.

WP1 · M1–M3

Measure the budget

n^* measured for all 20 sites under a fixed protocol — the ground-truth reference.

LARGELY COMPLETE

WP2 · M2–M9

Formalize & build the gate

Source and target as **relational roles**, the graded relations that mediate them, and the epistemic status of a verdict.

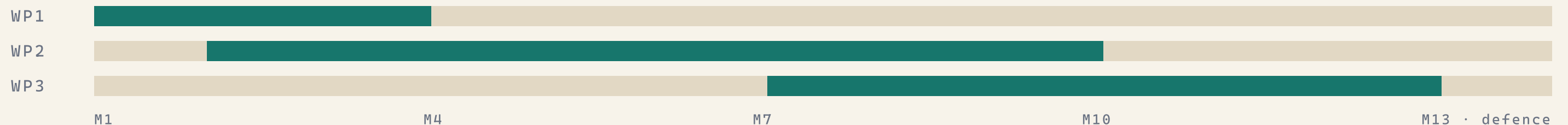
MOST DEMANDING PACKAGE

WP3 · M7–M12

Analyze against measured budgets

Test Ordering and Discrimination. The results **validate or falsify** the central claims.

AHEAD OF WRITE-UP



Transferability is a structured, investigable phenomenon — and formal ontology has a concrete, testable role in explaining it.

The same skeleton — a source model, a new context, a local adaptation budget, and competing similarity relations that may predict it — appears in hydrology, air quality, agriculture and almost every renewable source. The general contribution is a **way of asking**: state the ontological commitment, give it a role in a neurosymbolic system, and let the empirical outcome adjudicate.

PUSHBACK WANTED ON

How I individuate and operationalize the similarity relations.

PUSHBACK WANTED ON

The epistemic status I assign to a gate's verdict.

PUSHBACK WANTED ON

How to frame the ontology–learning interface so it speaks to both communities.

Rodrigo Huber Marques Moreira Mendes · rodrigo@aluno.puc-rio.br · ORCID 0009-0003-3001-2870 · PUC-Rio, Department of Industrial Engineering · Supervised by Fernanda Baião and Reinaldo Castro Souza · Supported by CAPES, CNPq and FAPERJ · Full statement in the FOIS 2026 ECS proceedings (CEUR-WS).